

Top 25 Data Science & SQL Interview Questions

Prepared for Bhaskar Mandal

SQL Interview Questions

1. Explain the difference between WHERE and HAVING.
2. What is the difference between INNER JOIN, LEFT JOIN, RIGHT JOIN, and FULL OUTER JOIN?
3. Write a SQL query to find the second highest salary.
4. Explain ROW_NUMBER(), RANK(), DENSE_RANK(), and NTILE().
5. Difference between GROUP BY and DISTINCT.
6. Explain Primary Key, Foreign Key, Unique Key, and Composite Key.
7. What is normalization? Explain 1NF, 2NF, and 3NF.
8. How do you find duplicate records in a table?
9. What are Common Table Expressions (CTEs)?
10. What is indexing? When can indexes slow queries?
11. Write a query to calculate a running total.
12. Find the top 3 highest-paid employees in each department.

Data Science Interview Questions

13. Explain one of your Data Science projects end-to-end.
14. Describe the complete Machine Learning pipeline.
15. How do you handle missing values?
16. Explain bias, variance, underfitting, and overfitting.
17. Which evaluation metrics do you use for classification, regression, and imbalanced datasets?
18. Explain Precision, Recall, F1 Score, and ROC-AUC.
19. Explain the Random Forest algorithm.
20. Compare XGBoost vs Random Forest.
21. How do you perform Feature Engineering?
22. What techniques do you use for Feature Selection?
23. How do you handle an imbalanced dataset?
24. Explain Bagging vs Boosting.
25. Based on your projects: Why did you choose the model? What challenges did you face? How would you improve it?